Equations of State for Realistic Water Models

1 Introduction

Water molecules are remarkably simple — consisting only of two hydrogen atoms and one oxygen atom — yet liquid water is highly complex, with well-known thermodynamic anomalies in its density,[1–4] heat capacity,[5–8] thermal conductivity,[9–11] diffusivity,[12–15] and surface tension,[16–19] amongst other properties. Many of these anomalies occur in either the supercooled, high-pressure, or negative-pressure regimes, which are experimentally difficult to access, yet have direct technological applications and consequences.[20–23] Nanoconfinement only amps up liquid water's bizarre array of oddities, such as strongly altered dielectric properties,[24–26] reduced friction,[27,28] interfacial-modulated reactivity,[29] and a potential superionic state.[30,31]

Molecular dynamics (MD) has become an increasingly popular technique to study liquid water systems, especially under nanoconfinement or at interfaces, for nanotechnological applications.[32–35] MD simulations are also naturally able to access experimentally-challenging regimes, such as deeply-supercooled liquid states and exotic high-pressure ices, without special difficulty.[36–39] In particular, the rise of machine learning potentials (MLPs) trained on *ab initio* reference data such as DFT allow highly accurate and realistic modelling of homogenous and heterogenous systems from first principles, capturing effects such as chemical reactivity and many-body interactivity without needing *post-hoc* adjustments.[40–45] MLP-driven MD simulations promise to accelerate materials design for technological applications involving water, by enabling key insights into water's behaviours at the nanoscale[46–48] and under extreme conditions.[4,30] Yet, conclusions drawn from MD simulations of liquid water, whether done through MLPs, computationally cheaper force fields (FFs), or even directly at an *ab initio* level, can only be as accurate as the underlying model.[49–53] This raises the critical question: how can we benchmark the performance and realism of different water models, especially across the extreme supercooled and high-pressure regimes?

In principle, the properties of liquid water under a given model can be measured using MD simulations across a wide range of thermodynamic conditions, and compared against experimental data pointwise. This is already a common practice, for example, when parametrizing FFs.[54–57] However, such a 'pointwise' approach is computationally wasteful: each state point requires a fresh simulation, and the resulting comparisons offer no general method of interpolating between sampled conditions, nor extrapolating towards extreme regimes. A more powerful

approach is to reconstruct the full Equation of State (EoS) of a given water model directly from a comparatively sparse set of simulations, by measuring fundamental thermodynamic quantities — such as the pressure and internal energy — as functions of volume and temperature. These can then be used to build a continuous representation of the model's free energy surface, allowing further properties of the model, at any state point within the domain of validity, to be obtained by direct evaluation without further simulation.

Our proposed approach offers two distinct advantages. Firstly, it allows the "realism" of a given water model to be assessed efficiently and comprehensively, by comparing its reconstructed EoS against experimental reference data across the full thermodynamic plane, rather than at a handful of isolated conditions. Secondly, and more importantly, it provides a natural framework for comparing models of foundationally different character — from simple, computationally cheap FFs such as SPC/E and TIP4P, to MLPs trained on high-accuracy DFT or coupled-cluster reference data — on equal footing, since the free energy surface construction makes no assumptions about the underlying interaction model, beyond requiring access to pressures and internal energies. This is particularly valuable in the supercooled and high-pressure regimes, where experimental measurements are difficult to obtain, but where accurate modelling is most needed to resolve long-standing questions about the microscopic origins of water's thermodynamic anomalies.

2 Theory

2.1 The Equation of State (EoS) in classical thermodynamics

Consider the canonical ensemble, i.e. a system of constant particle number N and volume V freely exchanging energy with a thermal reservoir of temperature T. The partition function is defined as:

$$Z(N,V,T) \;:=\; \frac{1}{{\mathcal{N}_0}^N}\int_{\boldsymbol{\omega}\in\Omega_N(V)} \exp\left[-\frac{1}{k_B T}\mathcal{H}(\boldsymbol{\omega})\right]\,\mathrm{d}^{6N}\boldsymbol{\omega} \tag{1}$$

where k_B is the Boltzmann constant, $\Omega_N(V)$ is the set of all available microstates at the given volume, $\mathcal{H}(\boldsymbol{\omega})$ is the Hamiltonian at microstate $\boldsymbol{\omega}$, and $\mathcal{N}_0$ is an arbitrary normalization constant which nondimensionalizes the expression. In general, the partition function is a smooth analytic function of V and T, except at phase transitions. The Helmholtz free energy is then defined as:

$$F(N,V,T) \;:=\; -k_B T \ln Z(N,V,T)\,. \tag{2}$$

Suppose there is some thermodynamic observable $A(\boldsymbol{\omega})$ which is a function of the microstate, and a corresponding conjugate variable λ such that $\partial_\lambda \mathcal{H} = A$. The thermal expectation of the observable is given by the Boltzmann distribution:

$$\langle A \rangle_{NVT} \quad = \quad \frac{\int_{\boldsymbol{\omega} \in \Omega_N(V)} A(\boldsymbol{\omega}) \exp\left[-\frac{1}{k_B T} \mathcal{H}(\boldsymbol{\omega})\right] \, \mathrm{d}^{6N}\boldsymbol{\omega}}{\int_{\boldsymbol{\omega} \in \Omega_N(V)} \exp\left[-\frac{1}{k_B T} \mathcal{H}(\boldsymbol{\omega})\right] \, \mathrm{d}^{6N}\boldsymbol{\omega}} \tag{3}$$

which, upon substituting eqs. (1) and (2), can be expressed in terms of the partition function and Helmholtz free energy respectively:

$$\langle A \rangle_{NVT} \quad = \quad -\frac{k_B T}{Z} \frac{\partial Z}{\partial \lambda} \quad = \quad \frac{\partial F}{\partial \lambda} \, . \tag{4}$$

As such, knowledge of the exact analytical form of the Helmholtz free energy $F(N, V, T)$ as a function of volume, temperature, and other parameters of interest λ is sufficient to derive all macroscopic equilibrium properties $\langle A \rangle_{NVT}$ in the canonical ensemble. Furthermore, by inverting the relevant variables, equilibrium properties in other ensembles can also be obtained; for example, the isobaric expectation $\langle A \rangle_{NpT}$ can be extracted by evaluating $\langle A \rangle_{NVT}$ at the inverse $V = V(p)$ of the $p(V)$ function. For this reason, especially where a single continuous thermodynamic phase of a given substance is concerned, the functional form of $F(N, V, T)$ is often known as the *Equation of State*.

2.2 EoSs for bulk phases

Now let us focus only on bulk condensed-phase substances, e.g. liquid water. In the thermodynamic limit $N \to \infty$, interfacial and edge effects vanish, and statistical fluctuations of bulk observables are suppressed relative to their expectation values. As such, all thermodynamic variables are either *extensive* and scale proportionally to N, or *intensive* and scale independently from N. It is useful to work only with intensive quantities; this can be achieved by dividing all extensive properties by the particle number N, or in other words we consider only quantities like the *volume per particle* or *internal energy per particle*, which for this subsection we mark with a circumflex:

$$\hat{V} \; := \; \frac{V}{N} \qquad ; \qquad \hat{U} \; := \; \frac{U}{N} \qquad ; \qquad \text{etc.} \tag{5}$$

In this chapter, we use the convention that each "particle" of liquid water refers to the entire water molecule, i.e. three atoms, or hydronium or hydroxide ions. This is a sensible choice, as the energy scale of the chemical bindings holding water molecules together far exceeds the thermodynamic energy scales which determine liquid water's bulk properties,[58–61] but any other convention would not affect results beyond a numerical factor.

Most importantly, we have the *free energy per particle*:

$$\hat{F}(\hat{V}, T) \quad := \quad \frac{F(N, V, T)}{N} \quad = \quad -\frac{k_B T}{N} \ln Z(N, V, T) \tag{6}$$

and eq. (4) can be applied without modification for intensive quantities if F is replaced by $\hat{F}$. For example, the pressure p and internal energy per particle $\hat{U}$ can be obtained from the derivatives along volume and temperature respectively:

$$p(\hat{V},T) = -\frac{\partial \hat{F}}{\partial \hat{V}}\,, \tag{7}$$

$$\hat{U}(\hat{V},T) = -T^2\,\frac{\partial}{\partial T}\left[\frac{\hat{F}}{T}\right]\,. \tag{8}$$

Other physically-meaningful derivatives are given in Table 1. The free energy per particle $\hat{F}$ therefore serves as a more useful Equation of State for bulk condensed-phase substances, where intensive properties are of greater interest than extensive ones due to trivial scaling with particle number. As such, for the rest of this chapter, we refer only to intensified quantities as per eq. (5) implicitly; that is, the circumflex symbol $\hat{}$, and the phrase "per particle", will not be used explicitly but should be kept in mind.

2.3 The dimensionless free energy

For the purposes of computer-assisted numerical optimization and fitting, it is often easier to work with dimensionless, purely numerical functions. A further simplification, as often performed in experimentally-fitted multiparameter EoSs,[62–66] can be made by factorizing $k_B T$ out from the free energy to leave a nondimensional quantity:

$$f := \frac{1}{k_B T}\,F = -\frac{1}{N}\ln Z \tag{9}$$

which we term the *dimensionless free energy*. Similarly, the volume and temperature variables can be nondimensionalized by dividing by some reference scale. In particular, the IAPWS-95[62] formulation, which serves as a standard reference for experimentally-fitted wide-range liquid and gaseous water EoSs, is expressed in terms of a *reduced density* and *inverse reduced temperature*:

$$\delta := \frac{V_c}{V}\,, \tag{10}$$

$$\tau := \frac{T_c}{T} \tag{11}$$

where $V_c = 92.9\,\text{Å}^3$ and $T_c = 647\,\text{K}$ are the volume and temperature at water's liquid-vapour critical point. The reason for this choice of variables is that it naturally bounds the low-pressure dilute-gas limit to a finite numerical range $0 < \delta, \tau < 1$, where the fluid behaves closely to an ideal gas and therefore does not require sophisticated description, while also expanding the high-pressure strongly-interacting dense-liquid regime to a much broader numerical range $1 < \delta, \tau < \infty$ which can be more easily adjusted by functional optimization. Scaling the

Table 1: **Physically-meaningful derivatives of the free energy per particle,** corresponding to macroscopic expectations of bulk thermodynamic observables in the canonical ensemble. All of these observables therefore can be calculated, if the form of $\hat{F}(\hat{V}, T)$ is known.

Name	**Symbol**	**Definition**	**Equivalent Identity**
Pressure	p	$-\left(\frac{\partial \hat{F}}{\partial \hat{V}}\right)_T$	
Entropy per particle	$\hat{S}$	$-\left(\frac{\partial \hat{F}}{\partial T}\right)_{\hat{V}}$	
Internal energy per particle	$\hat{U}$	$\frac{1}{N}\langle \mathcal{H} \rangle_{NVT}$	$\hat{F} + T\hat{S}$
Isochoric heat capacity per particle	$\hat{C}_V$	$\left(\frac{\partial \hat{U}}{\partial T}\right)_{\hat{V}}$	$-T\left(\frac{\partial^2 \hat{F}}{\partial T^2}\right)_{\hat{V}}$
Isobaric heat capacity per particle	$\hat{C}_p$	$\left(\frac{\partial \hat{H}}{\partial T}\right)_p$	$\hat{C}_V + \hat{V}T\beta_T{\gamma_v}^2$
Thermal expansion coefficient	α	$\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$	$\beta_T \gamma_v$
Isothermal compressibility	β_T	$-\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T$	$\left[\hat{V}\left(\frac{\partial^2 \hat{F}}{\partial \hat{V}^2}\right)_T\right]^{-1}$
Thermal pressure coefficient	γ_v	$\left(\frac{\partial p}{\partial T}\right)_V$	$-\frac{\partial^2 \hat{F}}{\partial \hat{V} \partial T}$
Isentropic bulk modulus	κ_s	$-V\left(\frac{\partial p}{\partial V}\right)_S$	$\frac{1}{\beta_T} + \frac{\hat{V}T{\gamma_v}^2}{\hat{C}_V}$

Table 2: **Dimensionally reduced thermodynamic observables from Table 1**, expressed in terms of the dimensionless free energy $f(\delta, \tau)$ as a function of reduced density $\delta = V_c/V$ and inverse reduced temperature $\tau = T_c/T$. Here subscripts denote partial derivatives, such as $f_\tau = \partial f/\partial \tau$.

Name	**Definition**	**Equivalent Identity**
Reduced pressure	$\dfrac{pV}{k_B T}$	δf_δ
Reduced entropy	$\dfrac{S}{k_B}$	$\tau f_\tau - f$
Reduced internal energy	$\dfrac{U}{k_B T}$	τf_τ
Reduced isochoric heat capacity	$\dfrac{C_V}{k_B}$	$-\tau^2 f_{\tau\tau}$
Reduced isobaric heat capacity	$\dfrac{C_p}{k_B}$	$-\tau^2 f_{\tau\tau} + \dfrac{\delta\,(f_\delta - \tau f_{\delta\tau})^2}{2f_\delta + \delta f_{\delta\delta}}$
Reduced thermal expansion coeff.	αT	$\dfrac{f_\delta - \tau f_{\delta\tau}}{2f_\delta + \delta f_{\delta\delta}}$
Reduced isothermal compressibility	$\dfrac{\beta_T k_B T}{V}$	$\left(2\delta f_\delta + \delta^2 f_{\delta\delta}\right)^{-1}$
Reduced thermal pressure coeff.	$\dfrac{\gamma_v V}{k_B}$	$\delta f_\delta - \delta\tau f_{\delta\tau}$
Reduced isentropic bulk modulus	$\dfrac{\kappa_s V}{k_B T}$	$2\delta f_\delta + \delta^2 f_{\delta\delta} - \dfrac{(\delta f_\delta - \delta\tau f_{\delta\tau})^2}{\tau^2 f_{\tau\tau}}$

variables based on V_c and T_c also fixes the critical point at $\delta = \tau = 1$, allowing non-analytic divergent terms to be inserted by hand at this specific point in the EoS.

Moreover, this choice of variables $f(\delta, \tau)$ simplifies many of the physically-meaningful free energy derivatives in Table 1, perhaps by coincidence. The pressure and internal energy from eqs. (7) and (8), for example, become:

$$\frac{pV}{k_B T} = \delta \frac{\partial f}{\partial \delta} , \tag{12}$$

$$\frac{U}{k_B T} = \tau \frac{\partial f}{\partial \tau} \tag{13}$$

where the LHS terms are known as the *reduced pressure* and *reduced internal energy*, noting their similarity to the expected expressions for the ideal gas. In fact, it can be easily shown — just from the above two equations — that the dimensionless free energy for the monatomic gas is given by $f(\delta, \tau) = \ln \delta + \frac{3}{2} \ln \tau$, showing the elegance of this representation.

Another particularly convenient derivative is the isochoric heat capacity:

$$\frac{C_V}{k_B} = -\tau^2 \frac{\partial^2 f}{\partial \tau^2} , \tag{14}$$

which we label the *reduced heat capacity*. Other derivatives are shown in Table 2. We thus proceed with developing liquid water EoSs using the $f(\delta, \tau)$ form.

3 Constructing the EoS

3.1 Problem statement

If both pressure and internal energy as functions of volume and temperature are known, then $f(\delta, \tau)$ can determined up to an additive constant by integrating eqs. (12) and (13). Fortunately, both of these are easily accessed in MD simulations. The EoS for a given water model can therefore be constructed by performing a sequence of canonical ensemble MD simulations of liquid water, fixed at varying volumes and temperatures, and obtaining a set of data points for $p(V, T)$ and $U(V, T)$; these measurements can then in principle be numerically integrated using e.g. the trapezoidal rule. In practice, however, a very fine grid of measurements is required to resolve curvatures in the integrand, resulting in prohibitively high cost for computationally expensive models. Furthermore, the sampled pressures and internal energies from each simulation are subject to statistical noise, and integrating them directly would result in an accumulation of correlated errors.

A preferable approach would be to instead construct a general flexible approximation or "model" for the dimensionless free energy function, by prescribing a fixed composition of elementary mathematical operations, and best-fitting the derivatives of the model onto eqs. (12)

and (13), and optionally also eq. (14), using standard regression techniques. The resulting best-fit model is therefore a targeted approximator for the true underlying $f(\delta, \tau)$, and is known as a *representation* for the Equation of State.

To be precise: suppose we have performed N_{sim} many simulations at the volumes $\{V_1, V_2, \ldots\}$ and temperatures $\{T_1, T_2, \ldots\}$, with the corresponding pressure results $\{p_1, p_2, \ldots\}$ and internal energy results $\{U_1, U_2, \ldots\}$. We wish to construct a model $\tilde{f}(\delta, \tau; \boldsymbol{\theta})$, which has some set of parameters $\boldsymbol{\theta} = \{\theta_1, \theta_2, \ldots\}$ controlling the behaviour of $\tilde{f}$. We may make $\tilde{f}(\delta, \tau; \boldsymbol{\theta})$ a good approximation of the true $f(\delta, \tau)$, by minimizing the *loss function*:

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}) &= \sum_{i=1}^{N_{\text{sim}}} \left| \nabla \tilde{f}(\delta_i, \tau_i; \boldsymbol{\theta}) - \nabla f(\delta_i, \tau_i) \right|^2 \\ &= \sum_{i=1}^{N_{\text{sim}}} \left[\left(\frac{\partial \tilde{f}(\delta_i, \tau_i; \boldsymbol{\theta})}{\partial \delta} - \frac{p_i V_i}{k_B T_i \delta_i} \right)^2 + \left(\frac{\partial \tilde{f}(\delta_i, \tau_i; \boldsymbol{\theta})}{\partial \tau} - \frac{U_i}{k_B T_i \tau_i} \right)^2 \right] \end{aligned} \tag{15}$$

so that the resulting best-fit $\tilde{f}(\delta, \tau; \boldsymbol{\theta}_{\text{min}})$ serves as a good representation of the EoS, and can be used to extract the thermodynamic observables of Table 2 efficiently and accurately. We may also optionally add heat capacity data $\{C_{V1}, C_{V2}, \ldots\}$ to the above loss function, as it relates to the second derivative $\partial^2 f / \partial \tau^2$.

The expressions of eqs. (1) and (9) guarantee that the target $f(\delta, \tau)$ is analytic almost everywhere within the phase space for a continuous state of matter, but the true function may be arbitrarily complex. The difficulty of modelling liquid water is shown, for example, by the plethora of functional forms and fitting parameters used in experimentally-fitted EoSs like IAPWS-95[62] and GERG-2008,[63] which employ 249 and 68 free fitting parameters respectively. Ultimately, there is no guarantee that a "simple" EoS can be constructed for any water model, especially if water's thermodynamic anomalies are to be captured faithfully. Simple polynomials, for example, may encounter Runge's phenomenon at high polynomial orders,[67,68] leading to unphysical model "blow-ups" even within the interpolation region. Experimentally-fitted EoSs choose a mixture of complex mathematical forms, such as logarithmic, exponential, and trigonometric functions, which are often hand-selected directly or indirectly,[62–66] but such ad-hoc models are inherently difficult to extend to new regimes, as shown by the variety of forms used in literature.

We therefore turn to a method inspired by the work of Brown (ref. 69), using local basis functions to build complete and stable approximators across arbitrary domains.

3.1.1 Local basis functions

For the context of this chapter, we define a family of C^k-smooth *local basis functions* to be a set of functions $\{B_1(x, y), B_2(x, y), \ldots\}$ satisfying:

- Each $B_i(x, y)$ is itself a C^k-smooth function everywhere on $\mathbb{R}^2$;

- For each $B_i(x, y)$, there is a finite simply-connected domain $\mathcal{D}_i \subset \mathbb{R}^2$ such that $B_i(x, y) = 0$ for all points outside $\mathcal{D}_i$;
- For any point (x, y), there exists at least one i such that (x, y) is in $\mathcal{D}_i$;
- The inner product is $\iint_{\mathbb{R}^2} B_i(x, y)B_j(x, y)\,\mathrm{d}x\,\mathrm{d}y = 0$ for all $i \neq j$;
- The norm $\iint_{\mathbb{R}^2} (B_i(x, y))^2\,\mathrm{d}x\,\mathrm{d}y$ is positive and finite (i.e. exists) for all i.

Note that the first two conditions also imply that $B_i(x, y)$, as well as all of its derivatives up to k^{th} order, go smoothly to zero at the boundary of $\mathcal{D}_i$. The first two conditions together thus mean that each B_i acts only as a localized "perturbation" within a small region of $\mathbb{R}^2$, while the third condition means that the domains of all the $\{B_i\}$ together "patch" the full space of inputs, and the fourth and fifth conditions allow the set of $\{B_i\}$ to form a complete orthonormal basis for the space of square-integrable functions.

We can therefore write any function as a linear combination of these local basis functions:

$$\tilde{f}(x, y; \boldsymbol{\theta}) \quad = \quad \sum_i c_i B_i(x, y) \tag{16}$$

where the coefficients $\{c_1, c_2, \ldots\}$ serve as the model parameters $\boldsymbol{\theta}$.

The use of local basis functions has a two-fold advantage. Since they form a complete orthonormal basis, the model is guaranteed to systematically converge towards the true underlying function as more terms are included in the expression of eq. (16), as long as there is sufficient data to calculate the coefficients c_i. More importantly, the use of localized functions allow independent additive perturbations to be made to the EoS representation; parameters in one thermodynamic regime need not change if new measurements in another regime require modification of the EoS. This makes the model *stable* against small perturbations of the fitting data.

Furthermore, the linear form of eq. (16) reduces the loss function in eq. (15) to a standard linear least-squares problem, which can be expressed as:

$$\mathcal{L}(\boldsymbol{\theta}) \;=\; \left| \begin{pmatrix} \partial_\delta B_1(\delta_1, \tau_1) & \partial_\delta B_2(\delta_1, \tau_1) & \partial_\delta B_3(\delta_1, \tau_1) & \ldots \\ \partial_\delta B_1(\delta_2, \tau_2) & \partial_\delta B_2(\delta_2, \tau_2) & \partial_\delta B_3(\delta_2, \tau_2) & \ldots \\ \vdots & \vdots & \vdots & \ddots \\ \partial_\tau B_1(\delta_1, \tau_1) & \partial_\tau B_2(\delta_1, \tau_1) & \partial_\tau B_3(\delta_1, \tau_1) & \ldots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix} - \begin{pmatrix} p_1 V_1 / k_B T_1 \\ p_2 V_2 / k_B T_2 \\ \vdots \\ U_1 / k_B T_1 \\ \vdots \end{pmatrix} \right|^2 \tag{17}$$

noting that the linear least squares problem $\mathcal{L}(\boldsymbol{\theta}) = |A\boldsymbol{\theta} - \mathbf{b}|^2$ has the well-known solution $\boldsymbol{\theta}_{\min} = (A^T A)^{-1} A^T \mathbf{b}$. This least-squares problem can also be further modified to include inverse-variance weighting, e.g. in the case of heteroscedastic data, or to propagate data uncertainties into parameter covariances.

3.1.2 Cardinal B-splines

The notation used in this section differs from the standard notation (e.g. as used in ref. 70). We begin by reviewing B-splines in the 1D case, before elaborating the 2D case.

A B-spline of order $p+1$ is a collection of local basis functions $B_i^{(p)}(x)$, which are each piecewise polynomials of degree p, defined by a sequence of m many "knots" demarcating the domains of each $B_i^{(p)}(x)$. We denote these knots as $\{t_1, t_2, \ldots, t_m\}$, which must be sorted into nondecreasing order $t_1 \leq t_2 \leq \ldots \leq t_m$. The B-spline is said to be "cardinal" if its knots are spaced equally; that is to say, the interval $\delta t = t_{i+1} - t_i$ is the same for all i. For our methodology, we use only cardinal B-splines.

B-splines may be constructed recursively via the Cox–de Boor formula. This starts with the $p = 0$ case, which are piecewise constant functions:

$$B_i^{(0)}(x) \;:=\; \begin{cases} 1 & \text{if } t_i \leq x < t_{i+1}\,, \\ 0 & \text{otherwise.} \end{cases} \tag{18}$$

The higher order B-splines are then defined by recursion:

$$B_i^{(p)}(x) \;:=\; \frac{x - t_i}{p\,\delta t}\, B_i^{(p-1)}(x) \;+\; \frac{t_{i+p+1} - x}{p\,\delta t}\, B_{i+1}^{(p-1)}(x)\,. \tag{19}$$

It can be checked that the resulting $B_i^{(p)}(x)$ is zero everywhere outside of $t_i \leq x < t_{i+p+1}$, and is C^{p-1}-smooth. It can also be checked that the set of $\{B_1^{(p)}, B_2^{(p)}, \ldots, B_{m-p-1}^{(p)}\}$ forms a complete orthonormal basis within the space of $t_1 \leq x < t_m$, and both the derivative and integral of a B-spline are themselves B-splines.

Visualizing these B-splines, especially from the expression of eq. (19), may be challenging; the author finds it easier to visualize B-splines using the fact that their derivatives follow a similar recursion:

$$\frac{\mathrm{d}}{\mathrm{d}x} B_i^{(p)}(x) \;=\; \frac{1}{\delta t}\, B_i^{(p-1)}(x) \;-\; \frac{1}{\delta t}\, B_{i+1}^{(p-1)}(x) \tag{20}$$

i.e. the derivative of $B_i^{(p)}$ is simply "two copies" of $B_i^{(p-1)}$ added together, with one copy translated rightwards by δt and inverted, and the value of $B_i^{(p)}$ is obtained by integrating its derivative. An example is shown in Figure 1.

B-splines are highly useful for constructing smooth approximations of single-variable functions $f(x)$ within the range of $t_1 \leq x < t_m$, as they form a complete basis, and feature the same general ease of representation and computation associated with simple polynomials, but do not suffer from Runge's phenomenon. The usual expression is to use one specific value of p, which is typically not very large, and sum over the indices i representing translations of the basis functions over the knots:

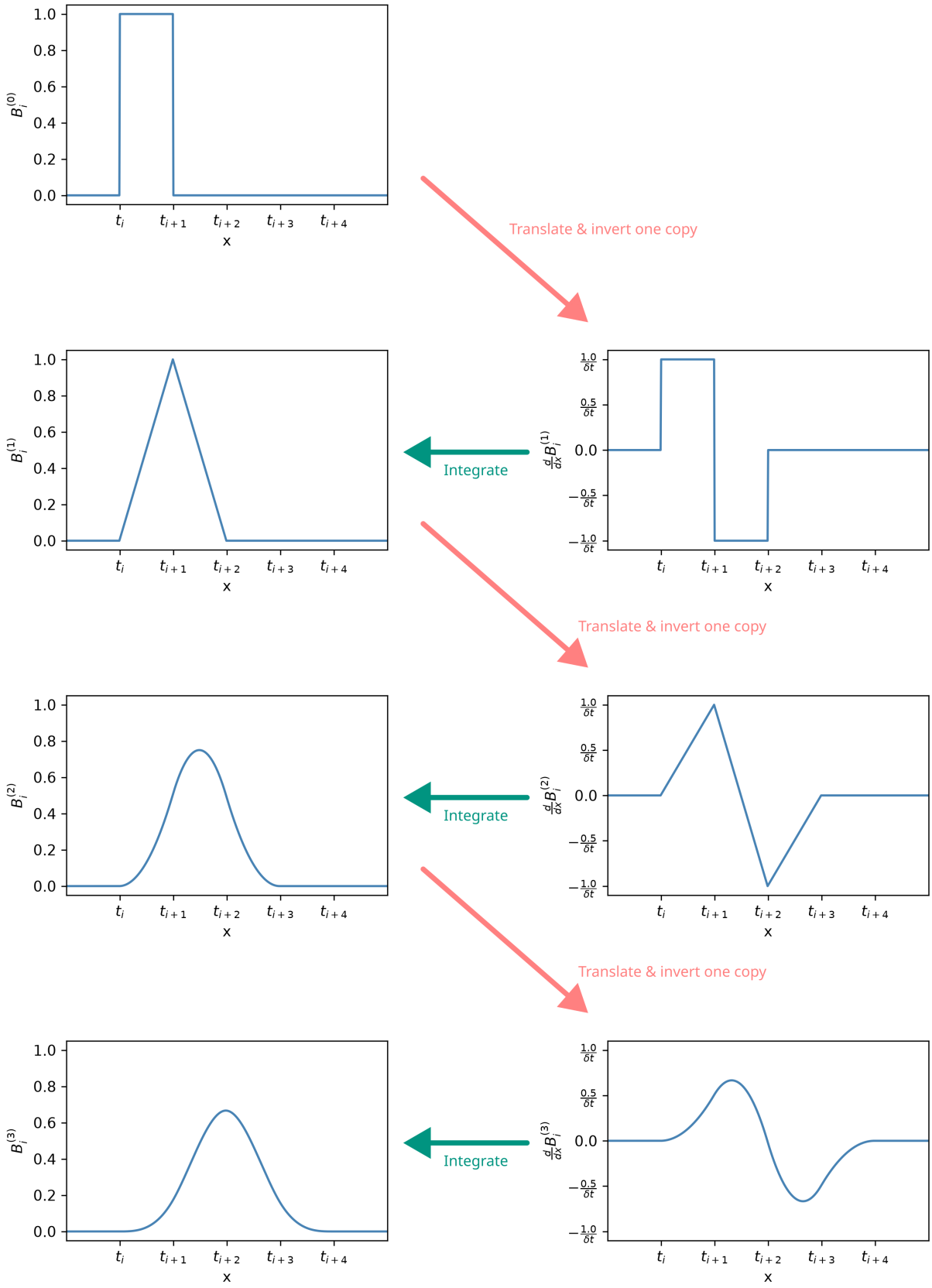


Figure 1: **Construction of 1D cardinal B-splines,** starting from $p = 0$ in the top row to $p = 3$ in the bottom row, using the derivative recursion of eq. (20).

$$f(x) \approx \sum_{i=1}^{m-p-1} c_i B_i^{(p)}(x) \tag{21}$$

with this being indicated as a "B-spline approximation of order $p+1$". Loosely speaking, the B-splines are able to approximate $f(x)$ by cutting the domain up into short intervals $t_i \leq x < t_{i+1}$, and fitting degree p polynomials independently within each interval with the constraint that all of the polynomials must connect smoothly. The local error of the approximation, at any point, is therefore on the order of $\mathcal{O}(\delta t^p)$. The finite truncation of the polynomials to a low degree p and short interval δt prevents spurious local divergences; as such, the accuracy of the approximation is usually improved by increasing the number of knots m, while leaving p small.

For our context, we may extend the theory of 1D B-splines into 2D by taking the tensor product of the basis functions over x and y:

$$\tilde{f}(x, y; \boldsymbol{\theta}) = \sum_{i=1}^{m_x-p_x-1} \sum_{j=1}^{m_y-p_y-1} c_{ij} B_i^{(p_x)}(x) B_j^{(p_y)}(y) \tag{22}$$

noting that the number of knots m_y, knot sequence $\{t_1, t_2, \ldots, t_{m_y}\}$, and order p_y along the y-axis may not necessarily be the same as for the x-axis.

3.1.3 Regularized B-splines for EoS models

For simplicity, we construct our EoS model using 2D cardinal B-splines with matching order $p_x = p_y = 4$. The choice of using degree 4 polynomials to represent the dimensionless free energy $f(\delta, \tau)$ ensures that its second derivatives, such as heat capacity, are piecewise quadratic. The knot sequences are determined by the input fitting data, chosen such that the endpoints t_1 and t_m are equal to the extremal values in the fitting data, and the total number of knot intervals in both dimensions is as close to the number of data points as possible. This means that the domain of (δ, τ) space covered by the model is the bounding box of the input fitting data, and is divided into subdomains spanned by smoothly-connected polynomials of degree 4.

This construction is already suitable for the linear least-squares form presented in eq. (17), if the tensorial coefficients c_{ij} are rearranged into a single vector of parameters $\boldsymbol{\theta}$. However, in practice the number of model parameters may exceed the number of independent data points — or, one or more subdomains may contain no data points, causing columns of zeros to occur in the coefficient matrix — leading to an underdetermined least-squares problem whose solution would be overfitted.

The solution to this is to introduce a *regularization*, which penalizes undesirable behaviours in the model fitting. In our case, we expect that the heat capacity, compressibility, and thermal pressure coefficient of any continuous phase of matter should be a smooth surface; these observables are all second derivatives of the free energy, hence a sensible regularization would be to minimize their curvatures by minimizing the fourth derivatives $\partial_\delta^4 \tilde{f}$ and $\partial_\tau^4 \tilde{f}$ of the model.

The choice of using B-spline order $p = 4$ guarantees that these fourth derivatives are piecewise constant within each subdomain, so there are an integer number of values of $\partial_\delta^4 \tilde{f}$ and $\partial_\tau^4 \tilde{f}$ which allows for easy construction of a *regularization matrix*.

Writing $\boldsymbol{\delta}$, $\boldsymbol{\tau}$, $\mathbf{V}$, $\mathbf{T}$, $\mathbf{p}$, $\mathbf{U}$ as the data vectors (i.e. observations from simulation data) of length N_{sim}, and $\boldsymbol{\theta}$ as the vector of model parameters, the regularized least-squares problem therefore has loss function given by:

$$\mathcal{L}(\boldsymbol{\theta}) \;=\; \left| \begin{pmatrix} \nabla_{\boldsymbol{\theta}}^{\mathrm{T}} \left[\partial_\delta \tilde{f}(\boldsymbol{\delta}, \boldsymbol{\tau}) \right] \\ \nabla_{\boldsymbol{\theta}}^{\mathrm{T}} \left[\partial_\tau \tilde{f}(\boldsymbol{\delta}, \boldsymbol{\tau}) \right] \\ \lambda \nabla_{\boldsymbol{\theta}}^{\mathrm{T}} \left[\partial_\delta^4 \tilde{f}(\boldsymbol{\delta}, \boldsymbol{\tau}) \right] \\ \lambda \nabla_{\boldsymbol{\theta}}^{\mathrm{T}} \left[\partial_\tau^4 \tilde{f}(\boldsymbol{\delta}, \boldsymbol{\tau}) \right] \end{pmatrix} \boldsymbol{\theta} \;-\; \begin{pmatrix} \mathbf{pV}/k_B\mathbf{T}\boldsymbol{\delta} \\ \mathbf{U}/k_B\mathbf{T}\boldsymbol{\tau} \\ \mathbf{0} \\ \mathbf{0} \end{pmatrix} \right| \tag{23}$$

where λ is a positive real number known as the *regularization parameter*; setting $\lambda \to 0$ returns the fitting problem to the unregularized case, whereas setting $\lambda \to \infty$ causes the model to prioritize "flattening" the curve everywhere instead of fitting the curve through the data points. A good choice of λ is to set it high enough such that the reduced χ^2 of the fit, i.e. fitting residuals normalized by data uncertainty, is on the order of unity. This can be achieved by performing a series of trial fits over varying λ, and selecting the fit with reduced χ^2 closest to one. A toy example in 1D is shown in Figure 2.

3.1.4 Summary: fitting EoSs

We choose to model the dimensionless free energy using cardinal B-splines in 2D, constructed by taking the tensor product of 1D B-splines of order $p = 4$:

$$\tilde{f}(\delta, \tau; \boldsymbol{\theta}) \;=\; \sum_{i=1}^{m_x - 5} \sum_{j=1}^{m_y - 5} c_{ij} B_i^{(4)}(\delta) B_j^{(4)}(\tau) \tag{24}$$

where the knot sequences are chosen to span the input fitting data's bounding box, with roughly as many intervals as data points. This model is then fitted to the input data using the regularized least-squares loss function given by eq. (23), with the regularization parameter λ chosen to bring the reduced χ^2 of the fit to unity.

In this way, a single set of simulation data, carried out at a finite number of volume and temperature state points V_i and T_i with corresponding results for pressure $p_i = p(V_i, T_i)$ and internal energy $U_i = U(V_i, T_i)$, and also optionally heat capacity $C_{V\,i} = C_V(V_i, T_i)$, allows for the construction of an approximate Equation of State $f(\delta, \tau)$, and therefore obtainment of all thermodynamic observables given in Table 2.

Uncertainty estimates can then be performed using *bootstrapping*, a generic technique which makes no assumption on the fitting parameters' covariances. Specifically, we may construct an

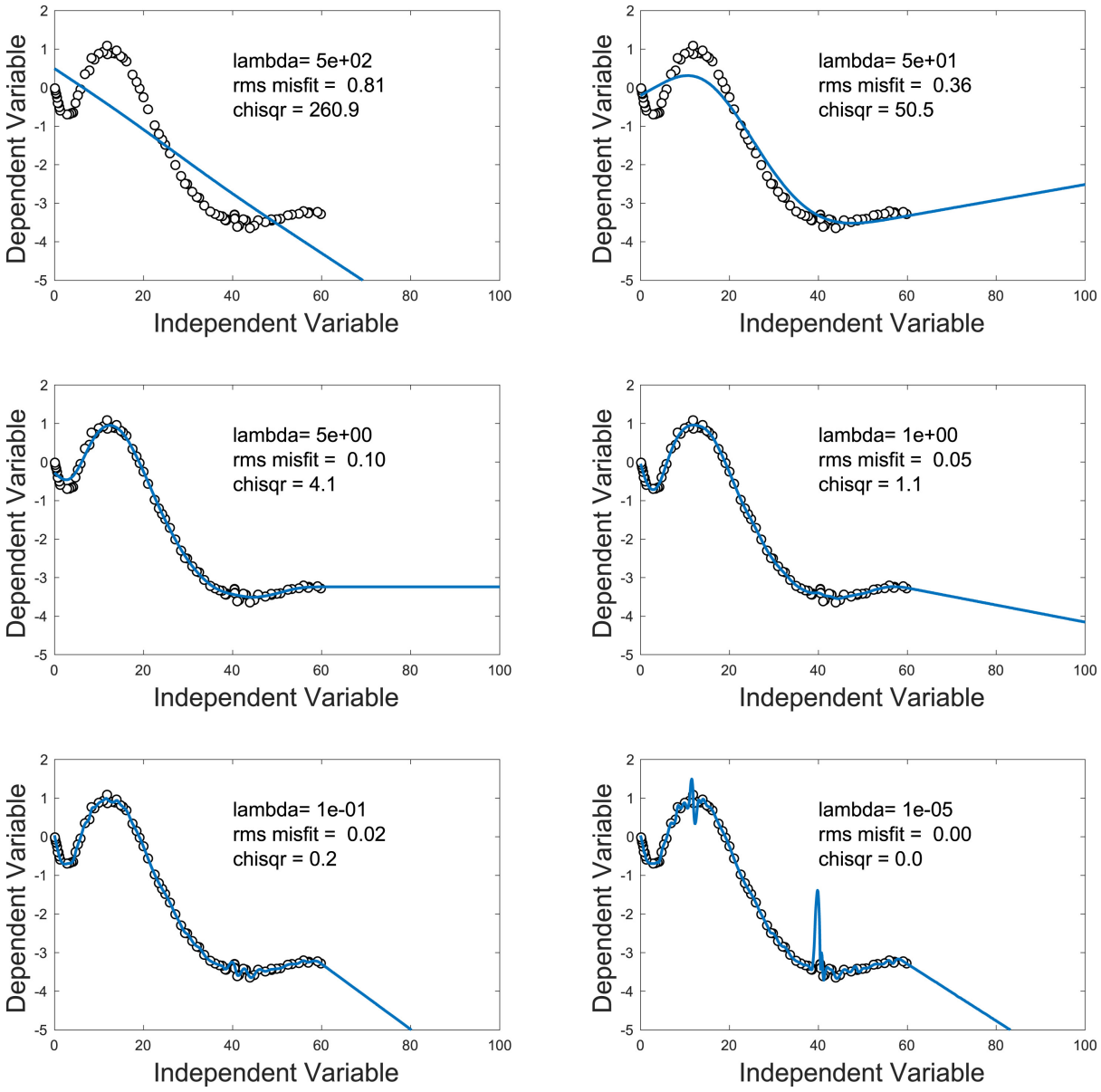


Figure 2: **Regularization of a 1D B-spline;** increasing the regularization parameter λ leads to a straight-line fit (top left), while decreasing λ leads to overfitting (bottom right). The best-fit is achieved when the reduced χ^2 is close to one (middle right). This figure is reproduced from ref. 69.

ensemble of EoS representations $\tilde{f}(\delta, \tau; \boldsymbol{\theta})$, where each model is fitted to randomly-perturbed copies of the original data with one point randomly removed. Hence, for any thermodynamic observable, we may "query" the entire ensemble of models and use the ensemble spread as a good estimator for the inherent uncertainty of the predicted observable.

4 Preliminary Results and Discussion

4.1 SPC/E liquid water

As a demonstration, we perform the EoS fitting procedure on liquid water as simulated using the SPC/E force field,[71] which models water molecules as rigid bodies of fixed O-H bond length 1.0 Å and fixed H-O-H angle 109.47°. The forces on the molecules are given by the electrostatic Coulomb interaction between hydrogen atoms of charge +0.4238 and oxygen atoms of charge −0.8476, as well as a Lennard–Jones interaction between oxygen atoms of lengthscale $\sigma = 3.166\,\text{Å}$ and energy scale $\varepsilon = 6.73\,\text{meV}$.

To be specific, a sequence of molecular dynamics simulations were performed, at a range of densities between $912.9\,\text{kg/m}^3$ to $1165.2\,\text{kg/m}^3$, and temperatures between 235 K to 370 K. These simulations were arranged as a uniform rectangular grid, targeting 6 values of density and 10 values of temperature, for a total of 60 simulations performed. Each simulation was carried out by creating a cubic box of 1,000 water molecules at the target volume, and running canonical ensemble molecular dynamics with timestep 1 fs at the target temperature using a Nosé–Hoover thermostat of relaxation time 0.5 ps for a simulation time of 10.5 ns (with the first 0.5 ns discarded for equilibration).

From each simulation, the instantaneous pressure $p(t)$ and total energy $E(t)$ is logged in 5 fs intervals. This allows extraction of the ensemble pressure, internal energy, and heat capacity as the time averages and fluctuation width respectively:

$$p(V,T) = \langle p(t) \rangle \,, \tag{25}$$

$$U(V,T) = \langle E(t) \rangle \,, \tag{26}$$

$$C_V(V,T) = \frac{\langle E(t)^2 \rangle - \langle E(t) \rangle^2}{k_B T^2} \tag{27}$$

and the corresponding uncertainties can be obtained from the statistical inefficiency-corrected variance, for the time averages, and from a statistical jackknife, for the fluctuation width.

The resulting EoS fits are shown in Figures 3 to 7.

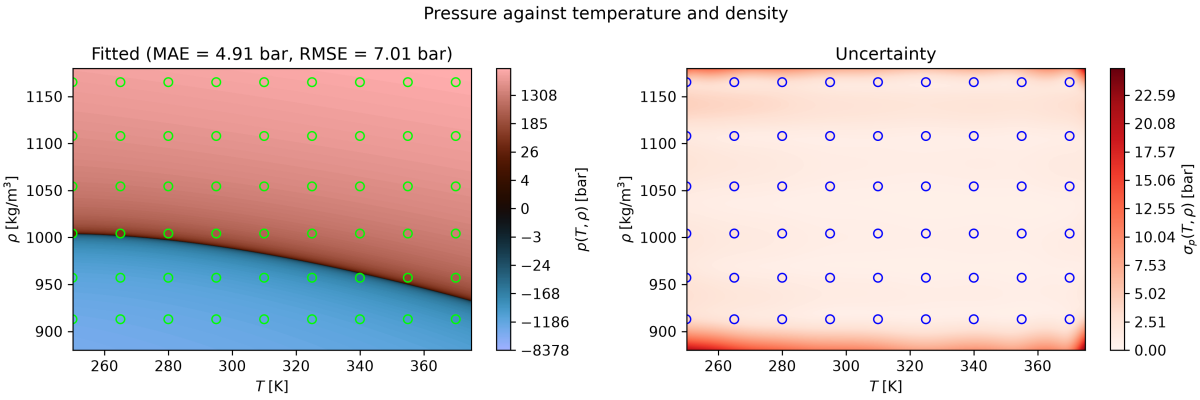


Figure 3: **Pressure as a function of density and temperature for the SPC/E model;** the plotted points are colored according to actual simulation data, while the background coloring is interpolated from the fitted EoS model $f(\delta, \tau)$.

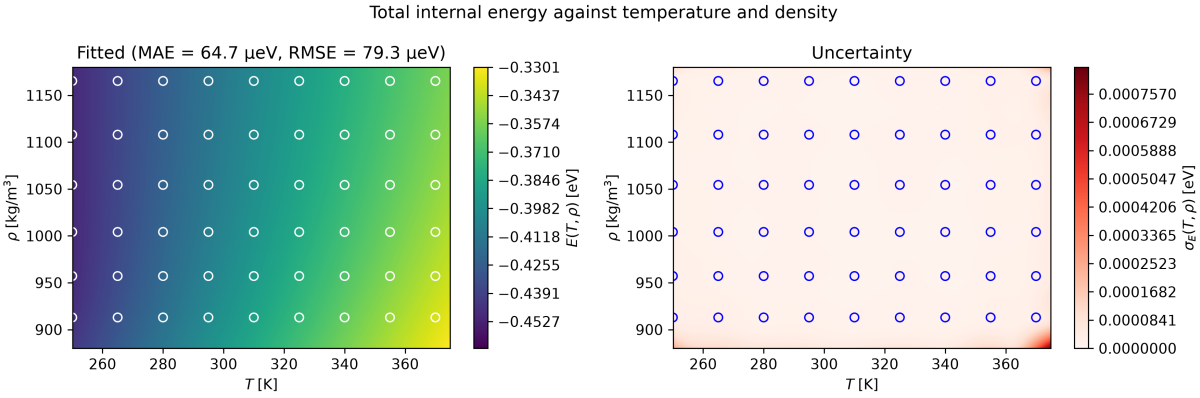


Figure 4: **Internal energy as a function of density and temperature for the SPC/E model;** the plotted points are colored according to actual simulation data, while the background coloring is interpolated from the fitted EoS model $f(\delta, \tau)$.

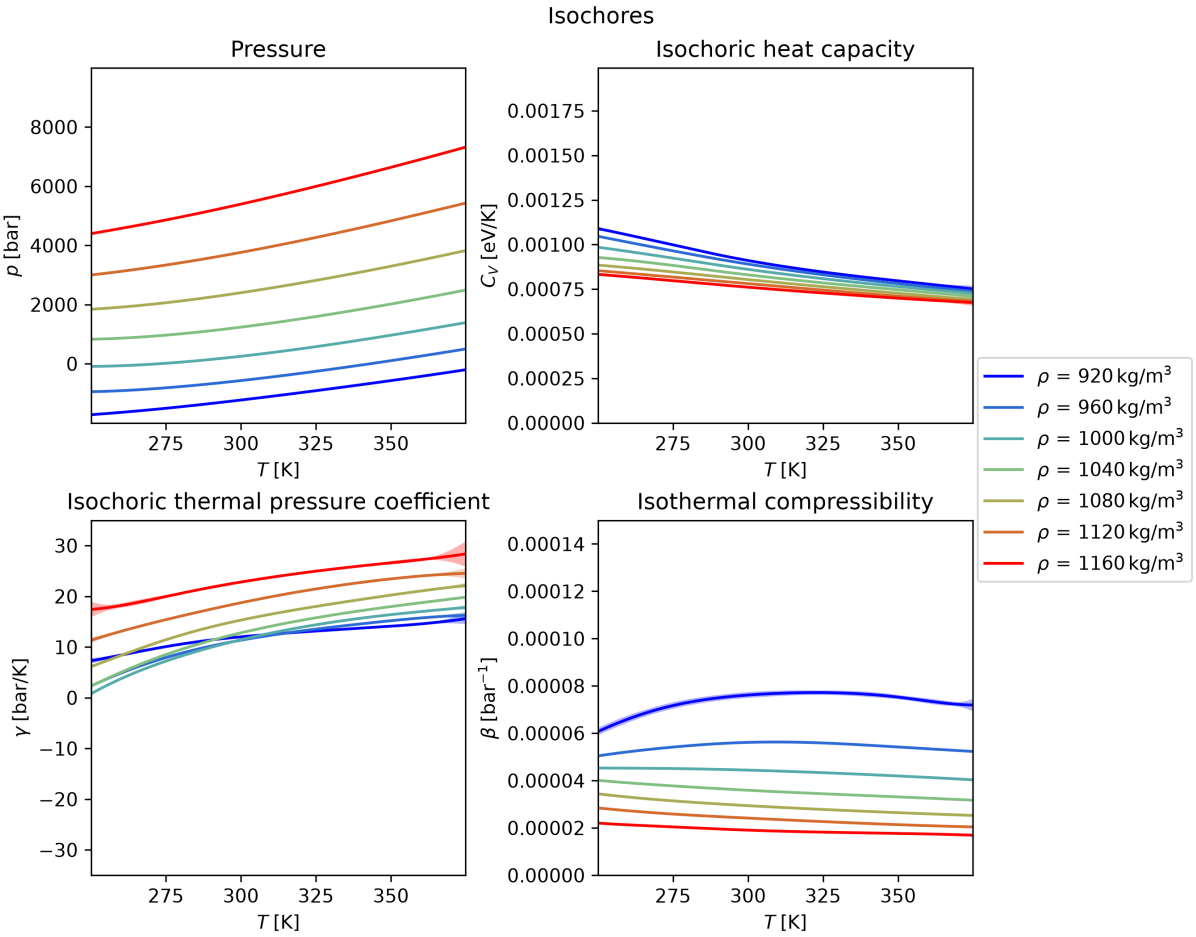


Figure 5: **Pressure, heat capacity, thermal pressure coefficient, and compressibility along isochores for the SPC/E model,** entirely obtained from the fitted EoS model $f(\delta, \tau)$. The lines are shaded according to the 95% confidence interval.

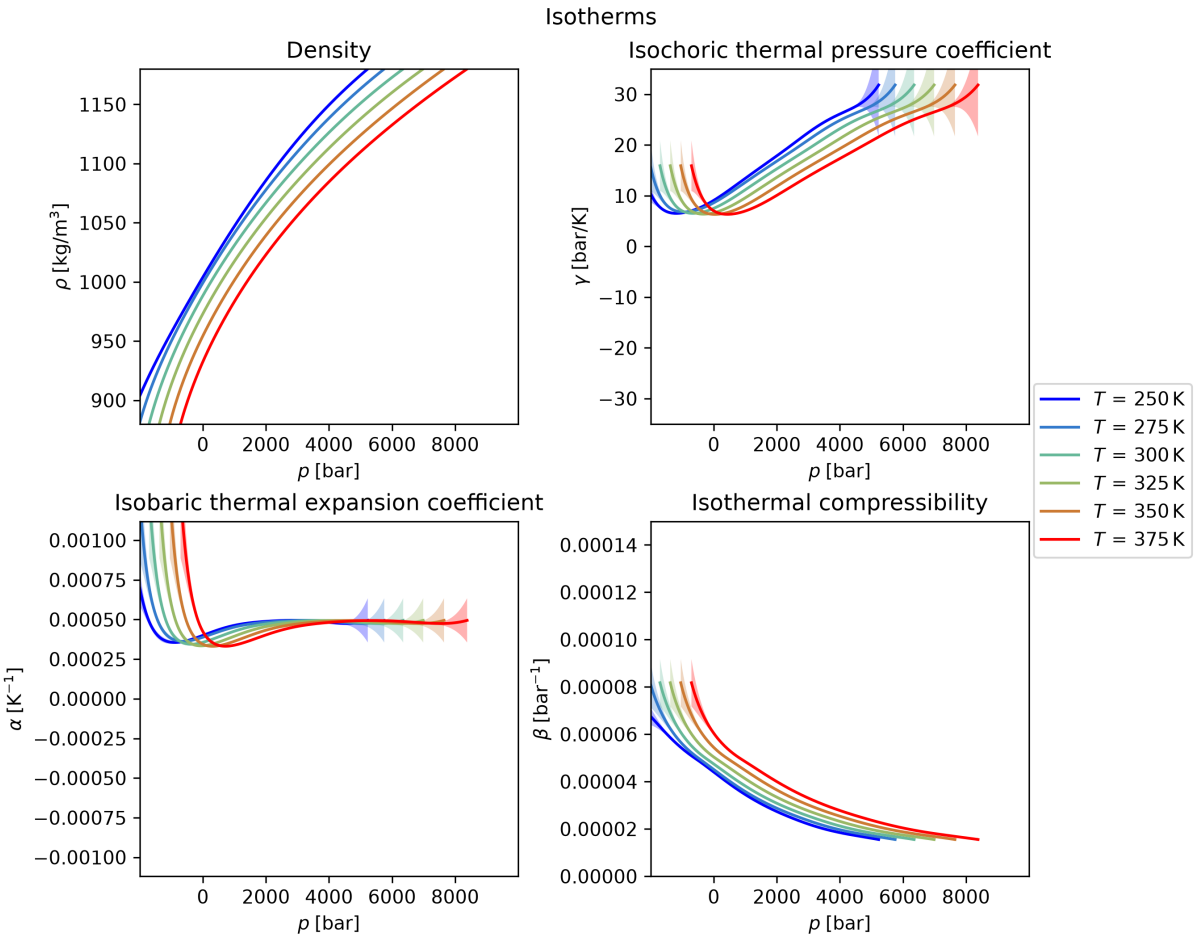


Figure 6: **Density, thermal pressure coefficient, thermal expansion coefficient, and compressibility along isotherms for the SPC/E model,** entirely obtained from the fitted EoS model $f(\delta, \tau)$. The lines are shaded according to the 95% confidence interval.

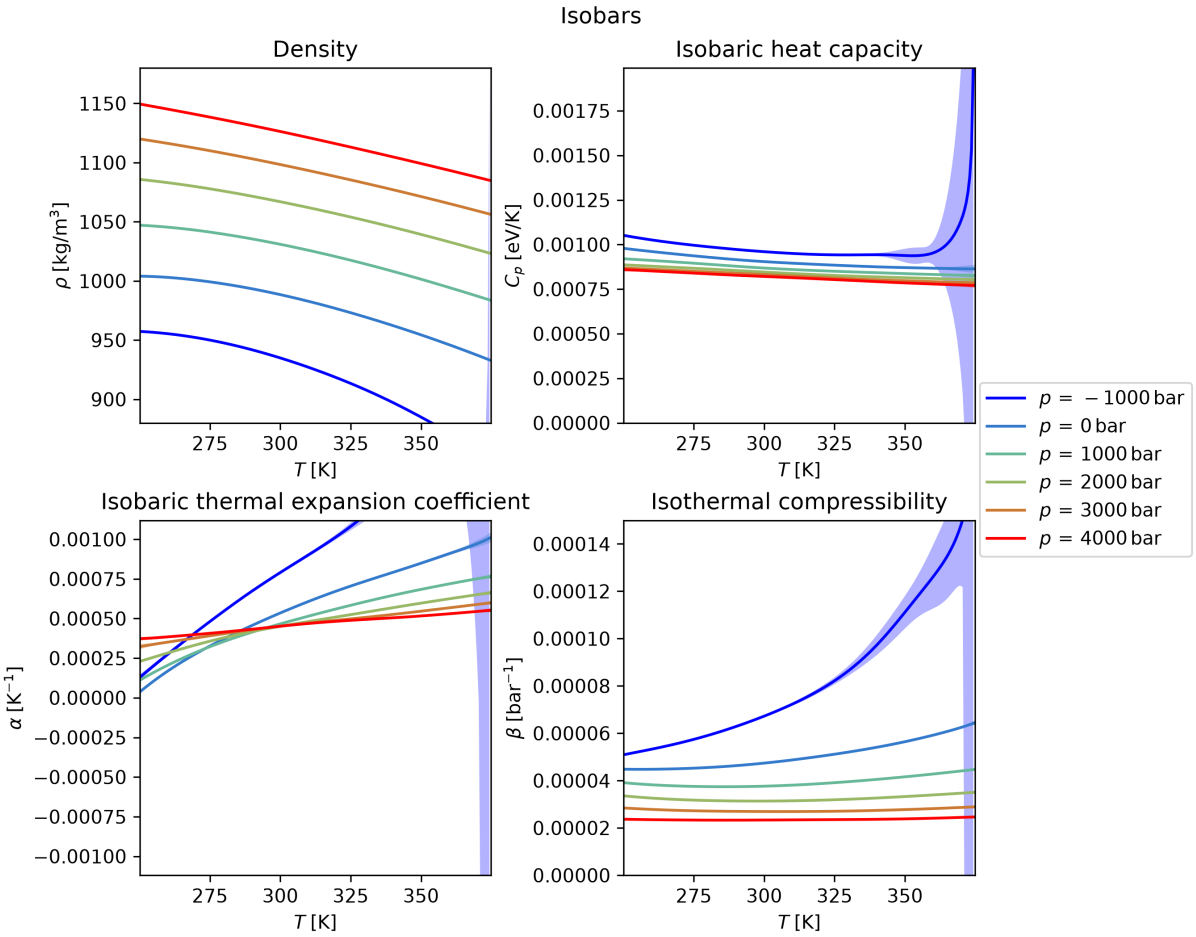


Figure 7: **Density, heat capacity, thermal expansion coefficient, and compressibility along isobars for the SPC/E model,** entirely obtained from the fitted EoS model $f(\delta, \tau)$. The lines are shaded according to the 95% confidence interval.

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